

MobileNet-v2 Compression on CIFAR-10: Uniform Quantisation and Sensitivity-Guided Mixed Precision

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CS6886 — System Engineering for Deep Learning, Assignment 2
Indian Institute of Technology Madras

1 Introduction

The deployed cost of a convolutional network is dominated by storage rather than arithmetic: parameters must be resident in memory, and activation tensors must be buffered between operators. Reducing the numerical width of both is the most direct lever on that footprint, and quantisation — representing tensors on a coarse integer grid instead of in 32-bit floating point — is the standard mechanism for doing so.

MobileNet-v2 [5] is a demanding subject for this treatment: its inverted residual blocks expand a narrow bottleneck, apply a depthwise 3×3 convolution and project back linearly, so the architecture is already parameter-efficient and carries little redundancy to absorb quantisation error. The evaluation workload is CIFAR-10 [3]: 50,000 training and 10,000 test images at 32×32 resolution over ten classes.

Weight precision and activation precision are separate budgets, and this report treats them as such throughout. Weights are quantised statically from the trained tensors and are stored once; activations are quantised against ranges estimated from calibration data and are re-materialised for every input. The experiments below establish an FP32 baseline (Q1), sweep both widths uniformly to locate the usable operating range (Q2), rank layers by second-order sensitivity to allocate precision non-uniformly (Q3), and audit the resulting storage against the uncompressed model (Q4).

2 Experimental Setup

Dataset and preprocessing. CIFAR-10 provides 50,000 training and 10,000 test images, 32×32 RGB, over ten classes. The training transform is `RandomCrop(32, padding=4)` with reflect padding, then `RandomHorizontalFlip(p=0.5)`, then `ToTensor` and `Normalize`; the horizontal flip is applied with probability $p = 0.5$. The test transform is `ToTensor` and `Normalize` only, with no stochastic operator, so evaluation is deterministic. Both splits use the channel statistics of the training set, $\mu = (0.4914, 0.4822, 0.4465)$ and $\sigma = (0.2470, 0.2435, 0.2616)$.

Model. The network is MobileNet-v2 with width multiplier 1.0 on $32 \times 32 \times 3$ inputs, reimplemented directly so that individual convolution and BatchNorm modules stay addressable by name for the compression stage. The published ImageNet schedule reduces resolution by $32\times$, which would drive a 32×32 input below one pixel before the classifier, so the stem convolution and the first expansion stage are set to stride 1. Total downsampling becomes $8\times$ and the classifier sees a $4 \times 4 \times 1280$ feature map. The inverted-residual stages (t, c, n, s) are otherwise unchanged:

$$\begin{aligned} &(1, 16, 1, 1) \quad (6, 24, 2, 1) \quad (6, 32, 3, 2) \quad (6, 64, 4, 2) \\ &(6, 96, 3, 1) \quad (6, 160, 3, 2) \quad (6, 320, 1, 1), \end{aligned}$$

with t the expansion factor, c the output channels, n the repeat count and s the stride of the first repeat. Activations are ReLU6, dropout is 0.2 before the classifier, and the model has 2,236,682 parameters. It is trained from scratch: transferring 224×224 pretrained weights to 32×32 inputs would require the very stride and resolution changes that discard most of the transferred benefit.

Training. Table 1 lists the recipe. Optimisation uses SGD with Nesterov momentum under a five-epoch linear warmup followed by cosine annealing [4] to zero, with label smoothing 0.1. A single seed (42) is threaded through model initialisation, shuffling and worker seeding.

Table 1: Training configuration (`src/train.py`).

Setting	Value	Setting	Value
Optimiser	SGD	Batch size	128
Momentum	0.9, Nesterov	Epochs	180
Weight decay	5×10^{-4}	Label smoothing	0.1
Learning rate	0.1	Seed	42
Warmup	5 epochs, linear	GPU	NVIDIA RTX A5000
Scheduler	cosine annealing	Epoch time	≈ 16 s

3 Q1 — Baseline Model and Evaluation

3.1 Baseline Configuration

The baseline is the model and recipe of the preceding section, trained for 180 epochs on one NVIDIA RTX A5000 at roughly 16s per epoch. The best test top-1 accuracy is **93.18%**, recorded at **epoch 176**. At the final epoch (180) the model reaches 98.70% train and 93.04% test accuracy; the ~ 5.7 pt gap is mild overfitting, consistent with a 2.2M-parameter model trained for 180 epochs on 50,000 images under light augmentation.

The epoch-176 best checkpoint (`runs/baseline/best.pt`) is used unchanged for every compression experiment in this report. No retraining, fine-tuning or quantisation-aware training is performed at any later stage, so all subsequent accuracy differences are attributable to quantisation alone.

3.2 Data Augmentation

Two stochastic transforms are applied at training time, resampled independently at every epoch so a given image is seen under a different realisation each pass. `RandomCrop(32, padding=4, reflect)` pads by four pixels per side and takes a random 32×32 window, translating the object within the frame; reflect padding avoids the black border that zero padding would introduce. `RandomHorizontalFlip(p=0.5)` mirrors the image with probability 0.5, which is label preserving for all ten CIFAR-10 classes, whereas a vertical flip is not and is therefore excluded. Figure 1 shows realisations of this pipeline.



Figure 1: Training-time augmentation. Each row shows one original CIFAR-10 image followed by four independent draws of `RandomCrop(32, padding=4, reflect)` composed with `RandomHorizontalFlip(p=0.5)`. Images are displayed after normalisation and inverse normalisation, so intensities match what the network receives.

3.3 Training and Test Curves

Figures 2 and 3 plot the recorded per-epoch history over the full 180-epoch schedule. Test accuracy oscillates through the high-learning-rate middle of the schedule, with brief dips around epochs 45–80, and smooths once cosine annealing brings the learning rate towards zero over the final ~ 30 epochs; the best test accuracy of 93.18% at epoch 176 is marked in Figure 3. The reported losses include label smoothing, so neither curve approaches zero at convergence.

3.4 Confusion Matrix

Figure 4 shows the confusion matrix of the epoch-176 checkpoint over all 10,000 test images, and Table 2 gives the corresponding per-class accuracies.

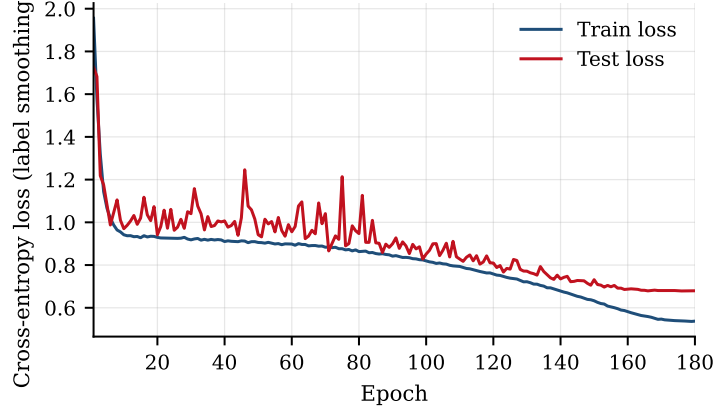


Figure 2: Training and test cross-entropy loss versus epoch, from `runs/baseline/history.csv`. Losses include label smoothing 0.1.

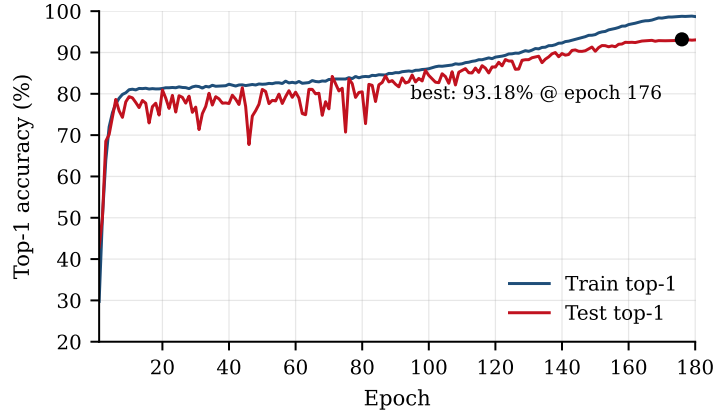


Figure 3: Training and test top-1 accuracy versus epoch. The marked point is the best test accuracy, 93.18% at epoch 176, which is the checkpoint used for all compression experiments.

The error is concentrated in one symmetric pair rather than spread across the matrix. Cat is the weakest class at 85.6%, with 6.8% of cats predicted as dogs; dog is the second weakest at 89.8%, with 6.2% predicted as cats. Those two cells alone account for roughly a fifth of the total classification error. The remaining off-diagonal structure is an order of magnitude weaker and semantically coherent — the two vehicle classes exchange some mass, as do ship and airplane — which is the standard CIFAR-10 failure pattern and not specific to this architecture.

Table 2: Per-class top-1 accuracy of the FP32 baseline (epoch-176 checkpoint) over the 10,000-image CIFAR-10 test set.

Class	Top-1	Class	Top-1	Class	Top-1
airplane	93.9%	deer	93.5%	ship	95.1%
automobile	96.7%	dog	89.8%	truck	96.2%
bird	91.3%	frog	95.2%	Overall	93.18%
cat	85.6%	horse	94.5%		

4 Q2 — Uniform Quantisation

4.1 Quantisation Method

Quantisation is symmetric and uniform with the zero point fixed at 0, which matches the roughly zero-centred weight distributions and removes the need to store an offset alongside each scale. For a tensor x quantised to b bits,

$$q_{\max} = 2^{b-1} - 1, \quad s = \frac{\max |x|}{q_{\max}}, \quad (1)$$

$$q = \text{clip}\left(\text{round}\left(\frac{x}{s}\right), -2^{b-1}, 2^{b-1} - 1\right), \quad \hat{x} = q s. \quad (2)$$

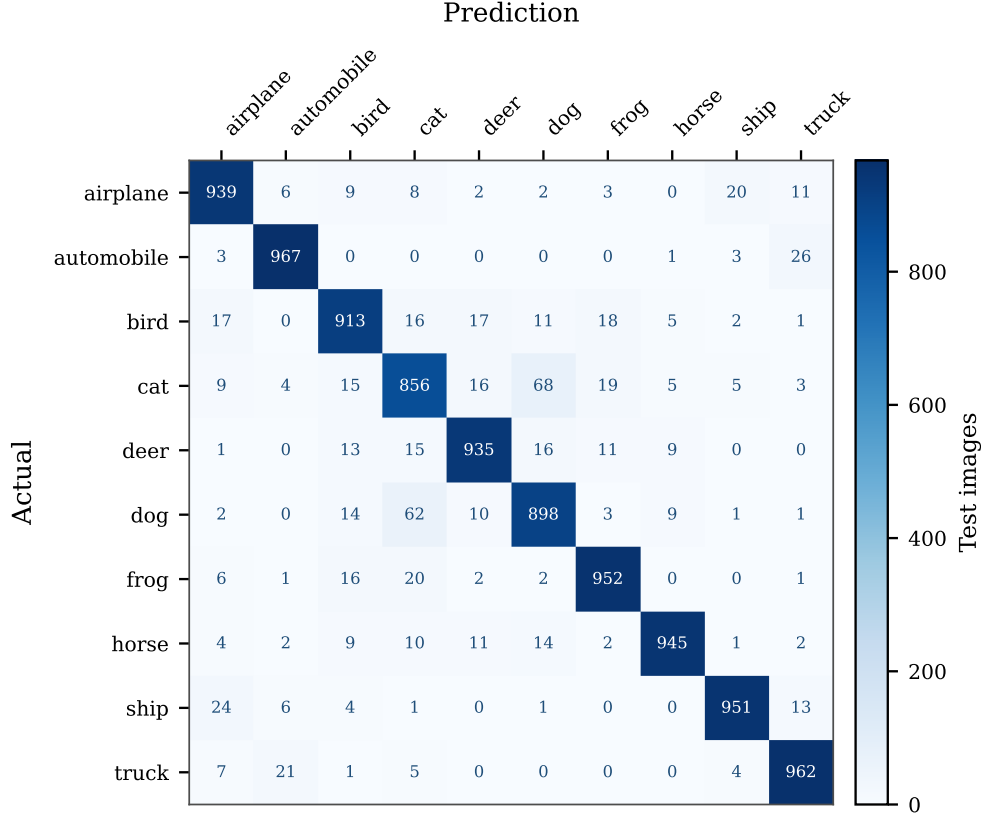


Figure 4: Confusion matrix of the FP32 baseline (epoch-176 checkpoint) over the CIFAR-10 test set. Cells give test-image counts. Each class contributes exactly 1,000 test images, so a diagonal entry divided by ten is that class’s accuracy in percent, matching Table 2.

Weights use per-output-channel scales computed statically from the trained FP32 tensors, so that one wide-range filter cannot set the grid for every other filter in its layer. Activations use one scale per layer input, obtained by static calibration: a running maximum of $|x|$ is accumulated over 512 CIFAR-10 training images with augmentation disabled, then frozen for all evaluation. Evaluation applies Eqs. (1)–(2) as fake quantisation, quantising and immediately dequantising each tensor, so the arithmetic error of the target precision is reproduced exactly. Storage uses explicit sub-byte packing: N values at b bits occupy $\lceil Nb/8 \rceil$ bytes, and every FP32 scale is charged its full four bytes.

All 53 Conv/Linear operators are quantised on both sides (Table 3). BatchNorm affine parameters and running statistics, all biases, the residual additions, the ReLU6 activations, pooling and the final logits remain in FP32; these tensors are numerically delicate relative to their size, totalling 68,234 elements or about 3% of the parameter count, and they are charged in full in the accounting of Section 6.3.

Table 3: Quantised operators. All 53 Conv/Linear layers are quantised on both the weight and the input-activation side.

Layer type	Count	Layer type	Count
Stem convolution (3×3)	1	Projection pointwise (1×1)	17
Expansion pointwise (1×1)	16	Final convolution (1×1)	1
Depthwise (3×3)	17	Classifier (linear)	1
Total quantised operators			53

4.2 Uniform Quantisation Results

Table 4 sweeps the two widths independently over $\{32, 8, 4, 2\}$ bits, and Figure 5 plots the same results against bit-width and against the achieved whole-model compression. The 32/32 row reproduces the FP32 accuracy exactly, confirming the wrapper is lossless and that every deviation below is caused by quantisation.

W8A8 retains almost all of the baseline accuracy, losing 0.40 pt. W4A8 remains usable at 90.75%, a 2.43 pt drop for an $8.00\times$ reduction of the weight tensors. W8A4, by contrast, collapses to 22.36% even

though its weights are the same 8-bit weights that score 92.78% at W8A8, and W4A4 behaves almost identically at 21.83%. Since weight width is held fixed across that pair, the failure is attributable to the activation side alone: under the max-absolute per-tensor calibration used here, activation precision is the binding constraint. Weight precision can therefore be reduced considerably more aggressively than activation precision, which is the observation the mixed-precision experiments in Section 5 exploit.

Table 4: Uniform quantisation sweep (runs/compression/). Compression ratios are tensor-level ratios for the quantised weights and for the per-image activation memory.

b_W/b_A	Top-1	Weight compression	Activation compression
32/32	93.18%	1.00×	1.00×
8/8	92.78%	4.00×	4.00×
4/8	90.75%	8.00×	4.00×
8/4	22.36%	4.00×	8.00×
4/4	21.83%	8.00×	8.00×
2/8	12.70%	16.00×	4.00×
2/2	10.00%	16.00×	16.00×

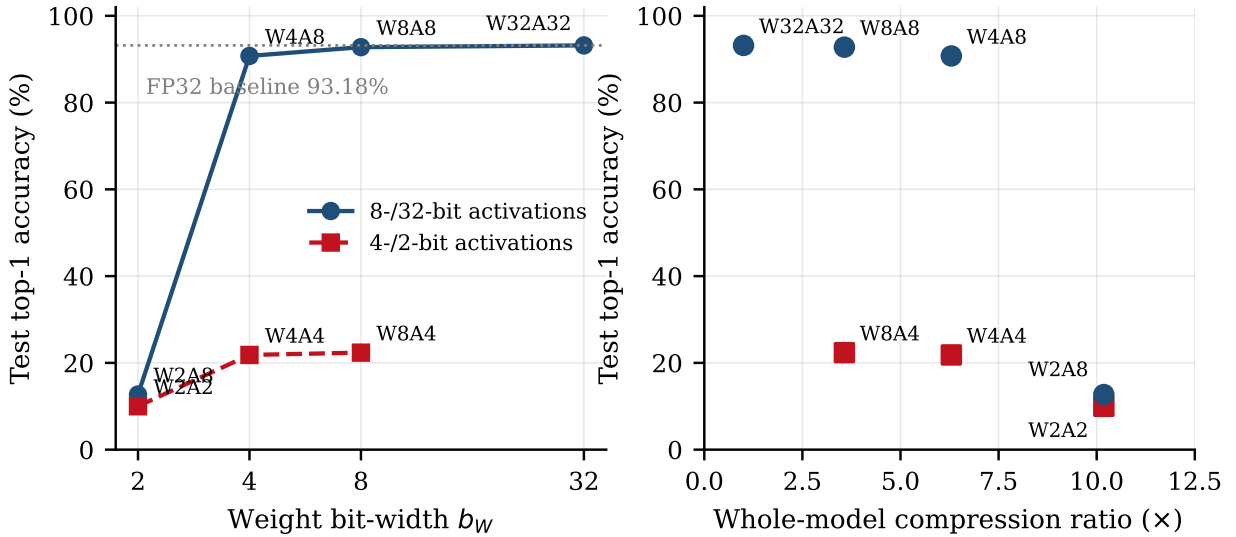


Figure 5: Uniform quantisation sweep. Left: test accuracy versus weight bit-width, separated by activation width. Right: accuracy versus achieved whole-model compression. Configurations with sub-8-bit activations sit below 25% irrespective of weight width.

5 Q3 — Sensitivity-Guided Mixed Precision

5.1 Layer Sensitivity

Uniform allocation implicitly assumes that every layer tolerates quantisation equally. HAWQ-V2 [1] replaces that assumption with a second-order criterion: a layer whose loss surface is sharply curved in its own weights suffers more from a given perturbation than a flat one. For layer l with n_l weights and Hessian block H_l , the sensitivity score combines average curvature with the perturbation that low-bit quantisation would actually inflict,

$$\Omega_l = \left| \frac{\text{Tr}(H_l)}{n_l} \right| \cdot \|W_l - Q_4(W_l)\|_2^2, \quad (3)$$

where $Q_4(\cdot)$ is the 4-bit quantiser of Eqs. (1)–(2).

The Hessian trace is estimated with Hutchinson’s method [2]: for a Rademacher random vector v , $\mathbb{E}[v^\top H v] = \text{Tr}(H)$, and each Hessian-vector product is obtained by double backpropagation through the loss. Estimates use 8 batches $\times$ 8 probes for each of the 53 Conv/Linear layers. The absolute value in Eq. (3) is deliberate: the trained model sits near but not exactly at a minimum, so a few layers return small negative trace estimates, and using the signed value would rank those layers as the safest to compress, inverting the intended ordering.

5.2 Sensitive Layers

Figure 6 ranks the twenty most sensitive layers. Sensitivity is concentrated in the early stages of the network: the seven highest-scoring layers are `features.1.conv.0.0`, `features.2.conv.1.0`, `features.2.conv.2`, `features.4.conv.2`, `features.1.conv.1`, `features.7.conv.2` and the stem `features.0.0`. Two properties explain the concentration: early layers hold few weights per output channel, so a coarse grid has little to average over and the perturbation $\|W_l - Q_4(W_l)\|_2^2$ is larger, and they feed every subsequent stage, so their error propagates through the full depth of the network. Later stages, which hold most of the parameters, score low and are the ones that can be pushed to 4 bits cheaply.

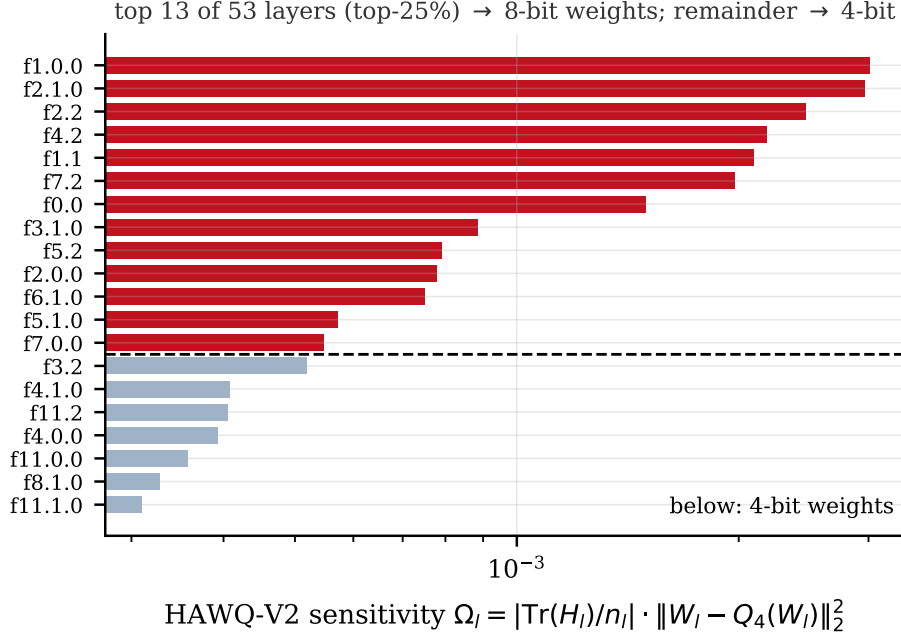


Figure 6: The twenty most sensitive of the 53 Conv/Linear layers, ranked by the HAWQ-V2 score of Eq. (3) on a logarithmic scale (layer names abbreviated, **f** = **features**). The thirteen layers above the dashed line form the top-25% set retained at 8-bit weights.

5.3 Mixed-Precision Results

Layers are sorted by Ω_l and the top fraction retains 8-bit weights while the remainder drops to 4 bits, with activations held at 8 bits. Table 5 reports the resulting sweep alongside the uniform baselines, and Figure 7 shows the corresponding Weights & Biases parallel-coordinates panel over weight width, activation width, compression ratio, model size and accuracy. The runs were logged offline, so the figure is the exported panel rather than a live dashboard.

Two comparisons carry the result. Against uniform 4-bit weights, `hawq_w8-4_top25_a8` gains 1.53 pt (92.28% versus 90.75%) at a comparable compression ratio. Against uniform 8-bit weights, `hawq_w8-4_top75_a8` is simultaneously more accurate (92.94% versus 92.78%) and more compressed (4.14× versus 3.74×), so the ranking is identifying genuinely different layer requirements rather than trading accuracy for size along a fixed curve.

Table 5: Mixed-precision sweep logged to Weights & Biases (`runs/mixed_precision/results.csv`). Compression is the whole-model ratio recorded for each run; the selected configuration is highlighted.

Configuration	Top-1 accuracy	Compression
<code>uniform_w32a32</code> (baseline)	93.18%	1.00×
<code>uniform_w8a8</code>	92.78%	3.74×
<code>uniform_w4a8</code>	90.75%	5.08×
<code>uniform_w4a4</code>	21.83%	6.91×
<code>uniform_w2a8</code>	12.70%	6.19×
<code>hawq_w8-4_top75_a8</code>	92.94%	4.14×
<code>hawq_w8-4_top50_a8</code>	92.69%	4.54×
<code>hawq_w8-4_top25_a8</code>	92.28%	5.05×
<code>hawq_w8-4_top10_a8</code>	91.56%	5.07×

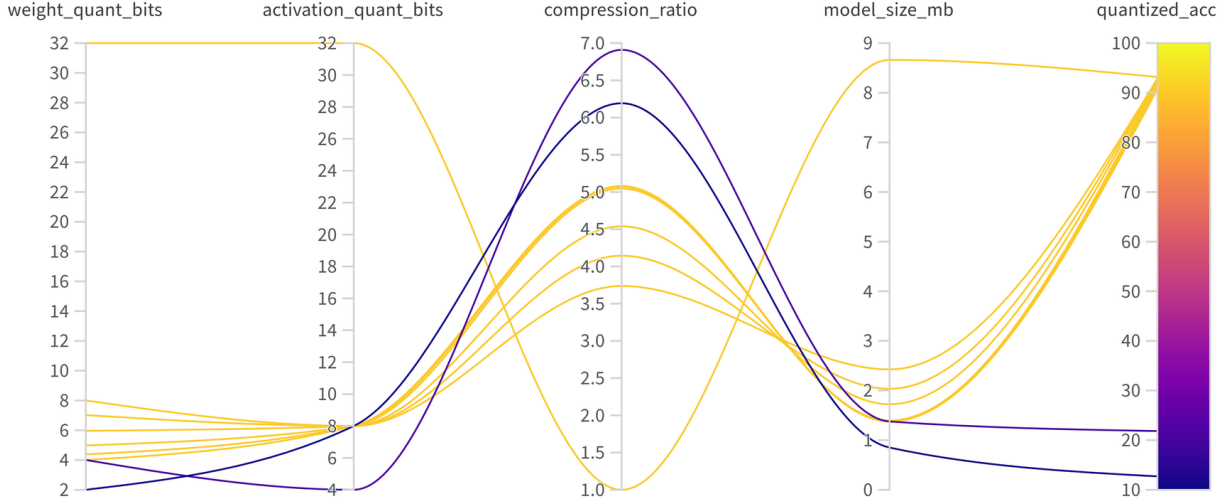


Figure 7: Weights & Biases parallel-coordinates panel over the configurations of Table 5, with axes for weight bit-width, activation bit-width, compression ratio, model size and quantised accuracy. Each polyline is one run, coloured by accuracy; runs were logged in offline mode.

5.4 Selected Configuration

The selected configuration is `hawq_w8-4_top25_a8`. The 13 most sensitive of the 53 Conv/Linear layers retain 8-bit weights, the remaining 40 use 4-bit weights, giving an average weight width of 4.07 bits, and all quantised activation inputs stay at 8 bits.

It sits at the knee of the accuracy/compression tradeoff: it reaches 92.28% top-1, 0.90 pt below the FP32 baseline, and improves on uniform W4A8 by 1.53 pt at approximately the same model size (1.395 MB versus 1.376 MB). Configurations that score higher cost substantially more memory for at most 0.7 pt, while everything more compressed degrades sharply.

6 Q4 — Discussion

6.1 Weight versus Activation Precision

The two sides of the network behave asymmetrically. Reducing weights to 4 bits while holding activations at 8 costs 2.43 pt (W4A8, 90.75%); reducing activations to 4 bits while holding weights at 8 costs more than 70 pt (W8A4, 22.36%). Because weight width is identical between W8A8 and W8A4, the collapse isolates cleanly to the activation path.

This asymmetry is a property of the calibration implemented here, not a general statement about low-bit activations. Scales are set per tensor from the maximum absolute value observed over the calibration set, so a single outlier fixes the grid for the whole tensor and the sixteen levels available at 4 bits are spread across the full observed range; the resulting error is then applied at each of the 53 quantised inputs. Alternative activation schemes — percentile or MSE-optimal clipping, finer granularity, or quantisation-aware fine-tuning — were not evaluated here.

6.2 Effect of Sensitivity-Guided Allocation

Non-uniform allocation is worth more than the equivalent uniform budget. At essentially the same size, `hawq_w8-4_top25_a8` reaches 92.28% against 90.75% for `uniform_w4a8`, a gain of 1.53 pt obtained purely by moving thirteen layers back to 8 bits. At the other end of the range, `hawq_w8-4_top75_a8` reaches 92.94% against 92.78% for `uniform_w8a8`, beating uniform 8-bit on accuracy while also being smaller. Spending the precision budget where curvature is highest therefore dominates spending it evenly.

6.3 Final Storage Accounting

Table 6 audits the selected configuration. The packed weights occupy 1.069 MB against 8.402 MB in FP32, a 7.86 $\times$ reduction. Activation memory, measured as the sum of the input tensors of the 53 quantised operators for a single image, falls from 6.194 MB to 1.549 MB, exactly 4.00 $\times$ at 8 bits. The complete model is 1.395 MB against 8.662 MB, a whole-model ratio of 6.21 $\times$.

The whole-model ratio is lower than the weight ratio because it charges for everything the model needs in order to run. Scale metadata contributes 68,476 B — one FP32 scale per output channel across

the quantised layers plus one per layer input — and the retained FP32 BatchNorm and bias tensors contribute 272,936 B. Together these are about a quarter of the compressed model, and neither shrinks when the weight width is reduced further.

Table 6: Final storage accounting for `hawq_w8-4_top25_a8`. The total model size includes 68,476 B of scale metadata and 272,936 B of retained FP32 BatchNorm and bias tensors.

Metric	FP32	Compressed	Ratio
Top-1 accuracy	93.18%	92.28%	−0.90 pt
Weight storage	8.402 MB	1.069 MB	7.86×
Activation memory / image	6.194 MB	1.549 MB	4.00×
Total model size	8.662 MB	1.395 MB	6.21×
Average weight width	32 bits	4.07 bits	—

6.4 Overall Interpretation

Sensitivity-guided mixed precision yields a better accuracy/compression tradeoff than uniform allocation because it retains higher precision only for the layers genuinely sensitive to quantisation, and those layers hold a small fraction of the parameters; uniform allocation must pay for precision across the entire network, including the late stages that tolerate 4 bits without measurable loss.

The accounting adds a systems-level qualification to that result. As the quantised weight payload shrinks, the fixed costs — scale metadata and the tensors deliberately retained in FP32 — come to dominate the model size. They are already 23.4% of the compressed model at 1.395 MB, which is why the whole-model compression curve flattens well before the weight-side curve does, and why further reductions in weight width would return progressively less. Any realistic model-size claim must include them.

7 Conclusion

A CIFAR-adapted MobileNet-v2 trained from scratch reaches 93.18% top-1 on CIFAR-10. Uniform quantisation of that baseline reveals a strong asymmetry between the two precision budgets: 4-bit weights with 8-bit activations remain usable at 90.75%, whereas sub-8-bit activations fail under the max-absolute per-tensor calibration implemented here, collapsing to 22.36% even with 8-bit weights retained. Ranking the 53 Conv/Linear layers by a HAWQ-V2-inspired sensitivity score improves the allocation of the weight budget, recovering accuracy at unchanged size.

The selected configuration, `hawq_w8-4_top25_a8`, assigns 8-bit weights to 13 layers and 4-bit weights to the remaining 40, with all quantised activations at 8 bits. It attains 92.28% top-1 accuracy at a final model size of 1.395 MB, corresponding to 7.86× weight compression, 4.00× activation compression and 6.21× whole-model compression including all quantisation metadata and retained FP32 tensors.

References

- [1] Zhen Dong, Zhewei Yao, Yaohui Cai, Daiyaan Arfeen, Amir Gholami, Michael W. Mahoney, and Kurt Keutzer. HAWQ-V2: Hessian aware trace-weighted quantization of neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 18518–18529, 2020.
- [2] Michael F. Hutchinson. A stochastic estimator of the trace of the influence matrix for laplacian smoothing splines. *Communications in Statistics – Simulation and Computation*, 18(3):1059–1076, 1989.
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